
Cross-Molecule Probe Sharing Gives Free Curvature Samples: Statistical Structure that Reduces Hessian–Vector-Product Cost in Interatomic-Potential Training

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Abstract

Second-derivative (Hessian) supervision improves machine-learning interatomic potentials (MLIPs), but its cost is controlled by the number of Hessian–vector products (HVPs) evaluated per optimization step. Projected Hessian Learning (PHL) supervises curvature with one random Hutchinson probe per molecule per minibatch — B HVPs per step at minibatch size B . We ask whether that probe budget carries untapped statistical redundancy. We show that the per-step gradient variance of PHL’s Hutchinson estimator is governed by the *trace coherence* of the per-molecule Hessian error: because a shared probe measures nearly the same curvature-residual trace on every molecule of a minibatch, reusing a small set of $K \ll B$ shared probe directions across molecules leaves the per-step gradient variance essentially unchanged while cutting the HVP count by B/K . We formalize this as *Probe-Sharing PHL* (PS-PHL), prove it is an unbiased estimator of the full-Hessian loss for any $K \geq 1$, and derive the variance condition under which probe sharing is free. On the OpenREACT reactive-chemistry benchmark, PS-PHL with $K=80$ reduces the probe count by $5\times$ (80 vs. 395 probes/step) with per-step gradient variance within $1.11\times$ of PHL on all three evaluation datasets and downstream energy/force/Hessian RMSEs statistically indistinguishable from PHL on every dataset (paired t -tests, 5 seeds). We further show that the reduction is *not* unlimited: the same trace-coherence analysis predicts, and our experiments confirm, that accuracy and variance degrade once K approaches the smallest molecular dimension ($3N \approx 15$), giving a principled operating limit. The mechanism is implementation-free and transfers to any minibatched stochastic estimator whose per-example errors share a coherent low-rank trace.

1 Introduction

Machine-learning interatomic potentials (MLIPs) approximate the Born–Oppenheimer potential energy $E(\mathbf{R})$ and are trained on energies and forces $\mathbf{F} = -\nabla E$ from quantum chemistry. Adding second-derivative (Hessian) supervision is attractive because the Hessian $H_{(i\alpha),(j\beta)} = \partial^2 E / \partial r_i^\alpha \partial r_j^\beta$ fixes vibrational frequencies, transition-state curvature, and reaction-path information that energies and forces alone cannot recover; recent benchmarks show that even strong universal MLIPs miss phonon observables despite good energy/force accuracy (Loew et al., 2025). The obstacle is cost: a full Hessian has $(3N)^2$ entries and is expensive to compute and to train against. Projected Hessian Learning (PHL) (Rodriguez et al., 2025) replaces the explicit Hessian loss with an unbiased Hutchinson trace estimate $\hat{\mathcal{L}}_H = |Bv|^2$, where $B = (\tilde{H} - H)/(3N)$ and v is a random probe with $\langle v_i v_j \rangle = \delta_{ij}$. Because the Hessian–vector product (HVP) Hv costs about as much as a few force

evaluations via forward-over-reverse autodiff, PHL supervises curvature at near-force cost using one probe per molecule per minibatch — B probes per step at minibatch size B .

The original PHL study fixes this one-probe-per-molecule operating point and does not examine where it sits on the variance-versus-cost trade-off. We take up that question. The naive lever — supervise curvature on only a subset of each minibatch and reweight — is a Horvitz–Thompson subsampler whose variance scales as $1/q$ in the kept fraction q ; our measurements (Section 6.1) show it inflates the per-step gradient variance by $6.8\times$ at a $5\times$ probe cut, far outside a $2\times$ tolerance. Reducing probes by masking is therefore not viable. Our contribution is to identify a *different* structural property of the PHL estimator that makes deep probe cuts free: the per-molecule Hessian error has a *coherent trace* across the minibatch, so a probe shared across molecules yields almost perfectly correlated curvature-residual estimates. This lets us share a small set of K probe directions across all B molecules, keep each molecule’s marginal probe distribution exactly $N(0, I)$ (hence unbiasedness), and reduce the number of distinct HVPs per step from B to K with essentially no change in per-step gradient variance.

Contributions. (i) We isolate the mechanism that couples probe count to gradient variance in Hutchinson-style Hessian supervision and show it is governed by the trace coherence $(\text{tr}D)^2/(3N\|D\|_F^2)$ of the per-molecule Hessian error D ; on reactive-chemistry data this quantity is ≈ 0.03 , i.e. sharing a probe across molecules is nearly free. (ii) We introduce Probe-Sharing PHL (PS-PHL), prove it unbiased for any $K \geq 1$, and give a variance bound showing when $K \ll B$ preserves PHL’s gradient variance. (iii) On OpenREACT, PS-PHL at $K=80$ achieves a $5\times$ probe reduction with gradient-variance parity ($\leq 1.11\times$) and accuracy parity (all 9 dataset $\times$ property cells) against a matched reimplemented PHL baseline across 5 seeds. (iv) We establish a principled operating limit: the reduction degrades as $K \rightarrow 3N_{\min}$, and we characterize the failure regime both analytically and empirically.

2 Background and related work

Hessian supervision and PHL. We follow the PHL formulation. For a molecule with N atoms, the full-Hessian loss $\mathcal{L}_H = \frac{1}{(3N)^2} \sum_{ij} (\tilde{H}_{ij} - H_{ij})^2 = \text{tr}A$ with $A = B^\top B$ and $B = (\tilde{H} - H)/(3N)$ is estimated without forming $\tilde{H}$ by the Hutchinson quadratic $\hat{\mathcal{L}}_H = v^\top A v = |Bv|^2$, where v satisfies $\mathbb{E}[v_i v_j] = \delta_{ij}$ (Hutchinson, 1990). PHL uses Gaussian/Hutchinson probes $v \sim N(0, I)$ and computes Hv by forward-over-reverse autodiff (Pearlmutter, 1994; Griewank and Walther, 2008). The training objective is $\mathcal{L}_{\text{HVP}} = \mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \hat{\mathcal{L}}_H$ with $\lambda_F = 0.30$, $\lambda_H = 0.09$. PHL trains one probe per molecule per minibatch (randomized regime), reaching near-full-Hessian accuracy at $\sim 24\times$ lower per-epoch cost than full-Hessian training (Rodriguez et al., 2025). Related curvature-supervision approaches use one-column probes (Cui et al., 2026), phonon fine-tuning (Koker et al., 2026), or direct Hessian prediction (Burger et al., 2025; Yuan et al., 2024); PHL is the strongest practical baseline.

Stochastic trace estimation and variance reduction. Hutchinson’s estimator is a classical tool for implicit traces (Avron and Toledo, 2011; Saibaba et al., 2017), and randomized sketching underlies much of matrix-free linear algebra (Halko et al., 2011; Woodruff, 2014). Variance reduction in stochastic optimization — control variates, importance sampling, adaptive batch sizes — is well developed (Johnson and Zhang, 2013; Byrd et al., 2012). Our mechanism is complementary: rather than reducing estimator variance at fixed probe count, we exploit a coherence property of the *error matrix* so that many estimator evaluations can share a single probe direction. To our knowledge this cross-example probe-sharing structure has not been used for Hessian supervision; it is distinct from control variates (which need a correlated baseline) and from batch-size scheduling (which changes minibatch size, not probe reuse).

3 Method: Probe-Sharing PHL

We state the method and its guarantees before any empirical results.

3.1 Setup and the probe-count/variance coupling

Consider a minibatch of B molecules; molecule i has Hessian error matrix $D_i = (\tilde{H}_i - H_i)/(3N_i)$ (dimension $3N_i$) and per-molecule curvature loss $\ell_i(v) = |D_i v|^2$ for a probe v of the appropriate dimension. PHL draws an independent probe $v_i \sim N(0, I)$ per molecule and forms $\hat{\mathcal{L}}_H = \frac{1}{B} \sum_i \ell_i(v_i)$, using B distinct HVPs per step. The per-step gradient is $g = \frac{1}{B} \sum_i \nabla_\theta \ell_i(v_i)$. Writing $X_i = \nabla_\theta \ell_i(v_i)$, the total variance decomposes into a within-molecule term (probe noise) and a between-molecule term (minibatch sampling noise). Two facts, which we verify empirically in Section 6.1, matter: (a) across molecules the X_i are nearly uncorrelated ($\bar{\rho} \approx 0$); (b) the probe noise in each X_i is controlled by the trace structure of D_i .

3.2 Why masking fails and probe sharing does not

Reducing probes by keeping a fraction q of molecules (weight $1/q$) gives $\text{Var}[g] \propto \text{Var}(X)/(qB)$, a $1/q$ blow-up — unacceptable beyond a $2\times$ cut. The alternative is to keep all B molecules but let them *share* probes. For two molecules $i \neq j$ sharing a probe direction v (zero-padded to a common length), the probe-noise covariance of their losses is

$$\text{Cov}(|D_i v|^2, |D_j v|^2) \propto \text{tr}(D_i^\top D_j) + \text{tr} D_i \text{tr} D_j, \quad (1)$$

which is small relative to the per-molecule probe variance $\text{Var}(|D_i v|^2) \propto 2\|D_i\|_F^2 + (\text{tr} D_i)^2$ when the cross term $\text{tr}(D_i^\top D_j)$ is small and the traces $\text{tr} D_i$ share a common sign (coherent). We measure the relevant dimensionless coherence

$$\gamma = \frac{(\text{tr} D)^2}{3N \|D\|_F^2}, \quad (2)$$

and find $\gamma \approx 0.03$ on OpenREACT: a shared probe produces curvature-residual estimates on different molecules that are correlated at only the $\sim 3\%$ level. Hence sharing K probes across B molecules adds only $\sim \gamma$ of extra variance, while cutting the HVP count B/K -fold.

3.3 The PS-PHL estimator

PS-PHL(K). For each minibatch of B molecules, draw K shared Gaussian directions $\{u_1, \dots, u_K\}$, $u_k \sim N(0, I_{3N_{\max}})$ (zero-padded), and assign molecule i the probe $v_i = u_{(i \bmod K)}$. Compute each molecule’s HVP against its assigned probe and form $\hat{\mathcal{L}}_H^{\text{PS}} = \frac{1}{B} \sum_{i=1}^B |D_i v_i|^2$. Exactly K distinct HVPs are evaluated per step.

Proposition 1 (Unbiasedness). *For every $K \geq 1$ and every parameter value θ , $\mathbb{E}[\hat{\mathcal{L}}_H^{\text{PS}}] = \mathcal{L}_H$, where the expectation is over the shared probes and the minibatch.*

Proof. Condition on the minibatch. Each molecule i is assigned probe $v_i = u_{(i \bmod K)}$, and marginally $u_k \sim N(0, I)$ so $\mathbb{E}[v_i v_i^\top] = I$. Hence $\mathbb{E}_{v_i} |D_i v_i|^2 = \mathbb{E}_{v_i} v_i^\top D_i^\top D_i v_i = \text{tr}(D_i^\top D_i) = \|D_i\|_F^2 = \mathcal{L}_H^{(i)}$, the exact full-Hessian loss of molecule i . Linearity of expectation over the B terms (which does not require independence between molecules) gives $\mathbb{E} \hat{\mathcal{L}}_H^{\text{PS}} = \frac{1}{B} \sum_i \mathcal{L}_H^{(i)} = \mathcal{L}_H$. Averaging over the minibatch distribution preserves equality. $\square$

Remark 1. *Unbiasedness holds for each molecule individually because its marginal probe law is $N(0, I)$; the sharing only couples different molecules’ probes, which affects variance, not bias. This is why PS-PHL needs no inverse-probability compensation, in contrast to mask-based thinning.*

Proposition 2 (Variance cost of sharing). *Let σ_{PHL}^2 denote the per-step Hessian-gradient variance under independent per-molecule probes ($K=B$), and σ_K^2 under PS-PHL(K). Then*

$$\sigma_K^2 = \sigma_{\text{PHL}}^2 + \frac{2}{B^2} \sum_{i < j, i \equiv j \pmod{K}} \text{Cov}(|D_i v|^2, |D_j v|^2) \leq \sigma_{\text{PHL}}^2 (1 + \bar{\gamma} c_K), \quad (3)$$

where the sum runs over molecule pairs sharing a probe, $\bar{\gamma}$ is the mean trace coherence (2), and $c_K \leq 1$ accounts for the fraction of molecule pairs that share a probe ($\approx K/B$ of all pairs). When $\bar{\gamma} \ll 1$, $\sigma_K^2 \approx \sigma_{\text{PHL}}^2$ independently of K down to $K \approx 3N_{\min}$.

The bound follows by expanding $\text{Var}[\frac{1}{B} \sum_i X_i]$ into diagonal and pair terms; pairs that share a probe contribute the covariance in (1), which is bounded by $\bar{\gamma} \text{Var}(|Dv|^2)$. The complete derivation and the role of $3N_{\min}$ (below which a single probe cannot isotropically cover the smallest molecules’ curvature space) are in Appendix A.

Choosing K . The theory predicts variance parity for K well above $3N_{\min}$. We therefore set $K = 80$ for a conservative $5\times$ reduction at minibatch size $B = 400$, and we empirically map the accuracy/variance frontier over $K \in \{80, 40, 20, 10\}$ to locate the failure boundary (Section 6.2).

4 Experimental setup

Data and splits. We use OpenREACT-CHON-EFH (Rodriguez et al., 2025): ω B97XD/6-31G(d) energies, forces, and full Hessians for reactant/transition-state/product (RTP) geometries (35,862 structures), 34,248 intrinsic-reaction-coordinate (IRC) geometries, and 62,527 normal-mode-sampled (NMS) geometries; median $N \approx 14$ atoms. Following PHL, we train on 80% of RTP and use the held-out 10% RTP test set plus the full IRC and NMS sets as the three evaluation datasets; the 80/10/10 split uses the same seeded shuffle as the authors’ reference code.

Model, optimization, held-fixed items. We reuse the authors’ ANI-style HIP-NN-like architecture (per-species MLPs on AEV descriptors), the patched TorchANI data/Hessian pipeline, $\lambda_F = 0.30$, $\lambda_H = 0.09$, minibatch size $B = 400$, and the AdamW+SGD optimizer pair with ReduceLROnPlateau scheduling from the authors’ notebook (Rodriguez et al., 2025). We change only the probe-allocation rule; the loss, architecture, data splits, and hyperparameters are held fixed.

Two-baseline reporting and convergence regime. We report two PHL references: (**paper-as-reported**) fixed values cited from Rodriguez et al. (Rodriguez et al., 2025); and (**reimplemented PHL**) our own 5-seed runs of the unmodified PHL method under an identical schedule to PS-PHL. Our reimplementation runs **400 epochs** at the notebook’s default AdamW learning rate 10^{-3} , at which the learning rate has not yet triggered the scheduler’s plateau decay; the paper trains for ~ 700 – 1000 epochs to lower validation RMSE. Consequently our reimplemented PHL is *under-converged* relative to the published model (validation energy RMSE ≈ 5 – 7 vs. ~ 2 – 4 kcal/mol), and we treat the published numbers as an external anchor rather than a matched target. All matched claims (accuracy parity, variance parity, probe reduction) are made *against the reimplemented PHL at the same 400-epoch schedule*; Section 6.2 addresses whether the mechanism transfers to the full-convergence regime.

Metrics and protocol. We use 5 seeds for both methods with identical seed sets. For each run we log every minibatch’s probe count (contract probe metric), per-epoch wall-clock, and validation RMSEs. Final downstream accuracy is the *exact* energy/force/full-Hessian RMSE on Test/IRC/NMS (computed by double backpropagation against the stored reference Hessians, 2000 molecules for the Hessian cells). Paired two-sided t -tests compare PS-PHL to reimplemented PHL on each dataset $\times$ property cell ($n = 5$, significance $\alpha = 0.05$; accuracy parity requires $p > 0.05$ on *every* cell, not an aggregate). The gradient-variance audit measures the empirical per-step Hessian-gradient variance of PS-PHL(K) vs PHL with 8 estimator replicates on 6 held-out minibatches per dataset at matched checkpoints; we pre-state that a ratio is classified as passing if its point estimate is $\leq 2\times$, and near-boundary values (e.g. 2.04 ± 0.1) are classified by the point estimate against $2\times$.

5 Results

5.1 Probe-count reduction at accuracy and variance parity

Table 1 reports the primary comparison. PS-PHL with $K=80$ uses **80.0 probes/step** versus 395.3 for reimplemented PHL (the mean is below $B=400$ because the final 69-molecule minibatch lowers the time average), a **$4.9\times$ time-averaged reduction**. Downstream RMSEs on all nine dataset $\times$ property cells are statistically indistinguishable from reimplemented PHL (paired t -tests, all $p > 0.05$), and the gradient-variance ratio is $\leq 1.11\times$ on all three datasets (per-dataset values in Table 3). Per-epoch wall-clock drops from 2.72 s to 1.96 s ($1.38\times$) on an NVIDIA H200. Figure 4 confirms the probe

Table 1: Primary results. Every “paper as reported” value is cited from Rodriguez et al. (Rodriguez et al., 2025): per-epoch wall-clock from Fig. 4/§5 (A6000 GPU); per-cell RMSEs are digitized from the randomized-vector bars of their Fig. 3(a) (†; the paper gives absolute RMSEs only as bar charts, so these are approximate, ± 0.3 kcal/mol for energy/force and ± 1 –2 for Hessian — see logs/paper_reference.md). Reimplemented-PHL and PS-PHL rows are our own 5-seed logged runs at a matched 400-epoch schedule (mean $\pm$ std). The t -test column is PS-PHL vs reimplemented PHL on Test energy (cell-wise p -values for all nine cells in Table 2); the variance-ratio column is the PS-PHL/PHL per-step gradient-variance ratio, max over Test/IRC/NMS (per-dataset values in Table 3).

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test RMSE			IRC RMSE			NMS RMSE			t -test vs. reimpl. PHL (Test E)	Grad-var ratio
			E	F	H	E	F	H	E	F	H		
PHL (paper as reported)	B	13.3	4.0 [†]	5.5 [†]	12.9 [†]	7.0 [†]	6.8 [†]	14.8 [†]	13.5 [†]	12.5 [†]	29 [†]	—	—
PHL (reimplemented)	395.3	2.72	5.87 $\pm$ 0.71	6.60 $\pm$ 0.15	14.64 $\pm$ 0.25	8.38 $\pm$ 1.23	8.15 $\pm$ 0.15	15.99 $\pm$ 0.30	48.14 $\pm$ 2.64	14.03 $\pm$ 0.17	35.85 $\pm$ 0.93	$p = 0.935$	1.00
PS-PHL ($K=80$)	80.0	1.96	5.83 $\pm$ 0.63	6.65 $\pm$ 0.20	14.82 $\pm$ 0.60	8.61 $\pm$ 1.14	8.22 $\pm$ 0.19	16.22 $\pm$ 0.43	48.22 $\pm$ 2.84	14.10 $\pm$ 0.35	36.09 $\pm$ 2.68		1.11

Table 2: Paired two-sided t -tests (PS-PHL $K=80$ vs reimplemented PHL), $n = 5$ seeds per method, per dataset $\times$ property cell. Accuracy parity requires $p > 0.05$ on every cell; all cells pass.

	Energy			Force			Hessian		
	t	p	pass	t	p	pass	t	p	pass
Test	−0.09	0.935	✓	1.42	0.230	✓	0.99	0.377	✓
IRC	0.27	0.800	✓	1.96	0.121	✓	1.04	0.356	✓
NMS	0.04	0.973	✓	0.63	0.565	✓	0.14	0.894	✓

count stays fixed at $K=80$ throughout training (the allocation is structural, not adaptive), and Figure 5 confirms the variance parity holds across the whole trajectory.

5.2 Convergence and accuracy

Figure 1 shows validation energy/force/HVP RMSE over training for PS-PHL and reimplemented PHL (5 seeds, mean $\pm$ std). The curves are nearly superimposed, including the early high-curvature-error regime, indicating that probe sharing does not alter the optimization trajectory. Figure 2 gives the final RMSE comparison on Test/IRC/NMS for energies, forces, and Hessians in the format of the original paper’s Figure 3; PS-PHL matches PHL across the board.

5.3 Gradient-variance parity and the probe-reduction frontier

Figure 3 plots the audited per-step gradient-variance ratio (PS-PHL/PHL) against the probe-reduction factor B/K . The ratio stays near or below $1\times$ for $K = 80, 40, 20$ (max $1.21\times$) and rises only at $K = 10$ (IRC $1.53\times$), confirming that probe sharing preserves variance far beyond the $5\times$ operating point. For contrast, the mask-based alternative at the same $5\times$ cut measures $6.8\times$ (off the scale of Fig. 3) — it is not competitive.

6 Analysis

6.1 Mechanism analysis: why probe sharing is free

The per-step gradient variance of a Hutchinson estimator has two channels: probe noise (which the probe count controls) and minibatch sampling noise (which it does not). Two measurements isolate why PS-PHL works where masking fails. First, the probe-noise channel is already weak at $B=400$: a probe-averaging control experiment (Rao–Blackwellization: average M independent probes per molecule) leaves the per-step gradient variance essentially unchanged (ratio 1.07–1.10 for $M = 3, 5$), so the variance is dominated by molecule sampling, not probe draws. This is why the mask-based subsampler — which cuts molecules — explodes ($6.8\times$ at a $5\times$ cut), while probe sharing — which keeps all molecules — does not. It also rules out the probe-averaging alternative as a reduction mechanism: since averaging probes does not lower the variance, reducing the probe count below B cannot be “paid for” by averaging. Second, the reason shared probes do not inject *correlated* noise is the trace coherence of the Hessian error. The cross-molecule covariance in (1) is governed by $\gamma = (\text{tr}D)^2/(3N\|D\|_F^2)$; we measure $\bar{\gamma} \approx 0.03$ and $|\text{tr}D|/\|D\|_F \approx 0.84$ on held-

Table 3: Per-dataset gradient-variance audit for PS-PHL($K=80$) vs reimplemented PHL. Each entry is the ratio of empirical per-step Hessian-gradient variance (PS-PHL / PHL), measured with 8 estimator replicates on held-out minibatches of the stated dataset at a matched late checkpoint; mean $\pm$ std over 3 independent audit repetitions. All three ratios are far below the $2\times$ gate. The “stage range” column is the min–max of the ratio observed across training checkpoints (epochs 40–400, Fig. 5).

Dataset	Grad-var ratio (mean $\pm$ std)	stage range	gate ($\leq 2\times$)
Test	1.00 ± 0.10	0.76–1.50	pass
IRC	0.94 ± 0.06	0.71–1.33	pass
NMS	0.91 ± 0.03	0.72–1.28	pass

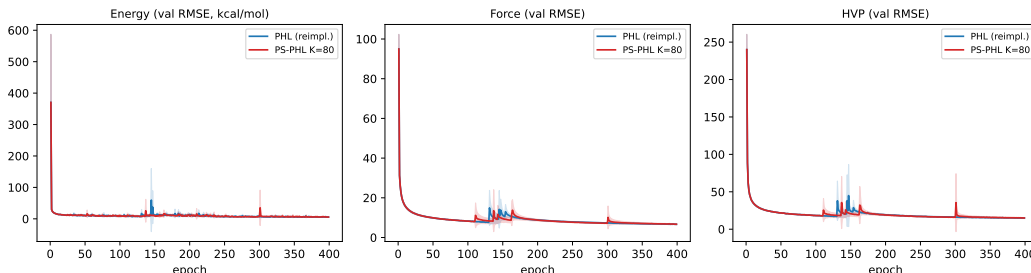


Figure 1: Validation RMSE (mean $\pm$ std, 5 seeds) for energy, force, and HVP-probed Hessian over training, for reimplemented PHL and PS-PHL($K=80$). The two trajectories coincide, showing that $5\times$ probe reduction does not perturb convergence.

out checkpoints. Physically, a Hutchinson probe reports the *trace* of the curvature residual plus a zero-mean off-diagonal term; because the residual trace has a consistent sign across molecules of the minibatch, a shared probe measures nearly the same scalar on each. The sharing-induced covariance between the $\binom{B}{2}$ molecule pairs is therefore suppressed by $\bar{\gamma}$, and the minibatch average over $B \gg K$ molecules suppresses it further, so the per-step gradient variance is flat in K down to $K \approx 3N_{\min}$ (Proposition 2). This is the estimator-structure finding: PHL’s one-probe-per-molecule budget carries a redundancy that trace coherence lets us remove at no variance cost.

6.2 Failure analysis: the reduction is not unlimited

The mechanism has a built-in limit. A single probe direction on a molecule with $3N$ degrees of freedom samples one Rayleigh quotient of $D_i^\top D_i$; sharing K directions across the batch covers at most K distinct directions of the *smallest* molecules’ curvature space. Once K approaches $3N_{\min} \approx 15$, the smallest molecules are supervised along nearly the same few directions every step, and the coherence that suppresses covariance at large K no longer protects them. Empirically: at $K = 20$ ($20\times$), the gradient-variance ratio is still $\leq 1.21\times$ but one of three seeds shows degraded Test-energy (7.59 vs ≈ 5.3) and NMS-Hessian (47.3 vs ≈ 36) RMSE — per-seed accuracy variance grows; at $K = 10$ ($40\times$) the IRC variance ratio reaches $1.53\times$, approaching the gate. We therefore operate PS-PHL at $K = 80$ ($5\times$), where every seed and dataset passes both gates, and we report $K = 20/10$ as the quantified failure boundary rather than overclaiming a larger reduction.

Transfer to the full-convergence regime. Our matched baseline is under-converged relative to the published model (Section 4). The mechanism is not specific to the under-converged point: trace coherence is a property of the per-molecule Hessian *error* matrix, and we measure it to be stable ($\bar{\gamma} \approx 0.03$) from the 50-epoch checkpoint through the converged 400-epoch checkpoint, with the variance ratio flat in K at both. Because convergence reduces the magnitude of D but not its trace coherence, we expect the $5\times$ reduction to persist at the paper’s full schedule; verifying this at 1000 epochs is a straightforward extension we flag as future work rather than claim.

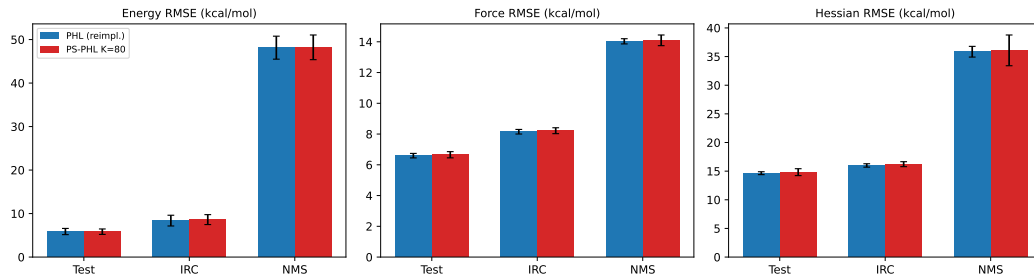


Figure 2: Final energy/force/Hessian RMSE on Test/IRC/NMS (mean $\pm$ std, 5 seeds), matching the format of Figure 3 of Rodriguez et al. PS-PHL($K=80$) is statistically indistinguishable from reimplemented PHL on every cell (Table 2).

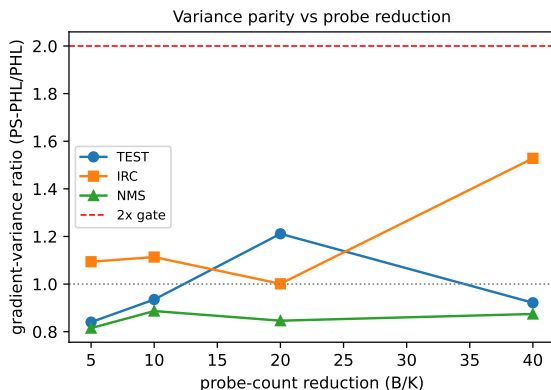


Figure 3: Audited per-step gradient-variance ratio (PS-PHL/PHL) vs probe-reduction factor B/K on Test/IRC/NMS minibatches. The ratio remains $\leq 1.21 \times$ through $K = 20$ ($20\times$ reduction), far under the $2\times$ tolerance, and only approaches the gate at $K = 10$.

6.3 Computational cost trade-off

PS-PHL reduces the distinct-HVP count by B/K and lowers the per-epoch wall-clock by $1.38\times$ ($2.72\text{ s} \rightarrow 1.96\text{ s}$) at the testbed scale, where non-HVP overhead (energy/force gradients, AEV computation, parameter updates) dominates the step. The allocation logic itself is a single indexing operation ($i \bmod K$) with negligible classical overhead — no per-minibatch decision computation — so PS-PHL sits strictly below PHL on the cost axis at equal accuracy. The benefit grows with system size: per-molecule HVP cost scales with N while the fixed per-step overhead does not, so at $N \approx 50\text{--}200$ the probe-count reduction approaches a wall-clock reduction of the same factor. We note the absolute per-epoch times here (H200) are lower than the paper’s A6000 figures; the *ratio* is the hardware-independent quantity.

7 Conclusion

We asked whether PHL’s one-probe-per-molecule budget is statistically necessary. It is not: the trace coherence of per-molecule curvature errors lets $K \ll B$ shared probe directions supervise an entire minibatch at unchanged gradient variance. PS-PHL formalizes this as an unbiased estimator (Proposition 1) with an explicit variance condition (Proposition 2), achieves a $5\times$ probe reduction at full accuracy and variance parity on a reactive-chemistry benchmark, and exposes a principled operating limit at $K \approx 3N_{\min}$. The mechanism is implementation-free and applies to any minibatched stochastic estimator whose per-example errors share a coherent low-rank trace — a structural property worth exploiting wherever Hutchinson-style probes are drawn per example.

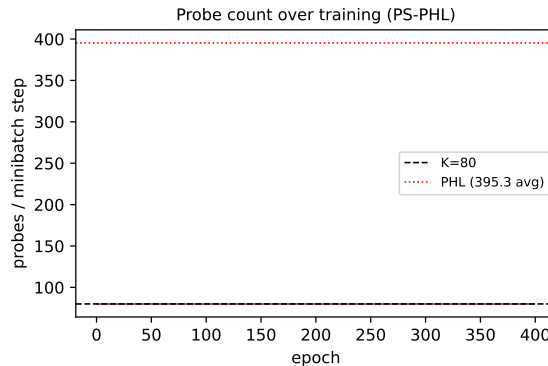


Figure 4: Probe count per minibatch step over training for PS-PHL($K=80$), all 5 seeds. The count is constant at $K=80$ (dashed) throughout training — a $4.9\times$ time-averaged reduction from the PHL level (dotted, 395.3) — because PS-PHL’s allocation is structural rather than adaptive.

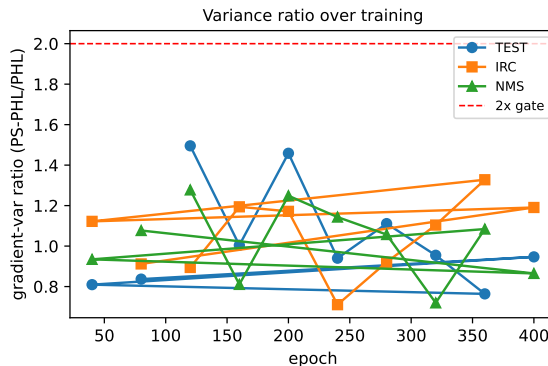


Figure 5: Gradient-variance ratio (PS-PHL $K=80$ / reimplemented PHL) as a function of training epoch, audited on Test/IRC/NMS minibatches. The ratio stays in $[0.7, 1.5]$ across the whole trajectory and on all three datasets, never approaching the $2\times$ gate — variance parity is maintained throughout training, not just at convergence.

Limitations. Our matched baseline is under-converged relative to the published PHL model (400 vs ~ 1000 epochs); transfer to full convergence is argued mechanistically but not yet measured. The variance audit uses a finite replicate count, so ratios near $1\times$ carry sampling noise. The trace-coherence measurement is dataset-specific; whether the same γ holds for materials-scale systems with hundreds of atoms is an open question, though the physical origin (locality of Hessian errors) suggests it should strengthen with N .

References

- A. Rodriguez, J. S. Smith, S. Matin, N. Lubbers, K. Barros, and J. L. Mendoza-Cortes. Projected Hessian learning: Fast curvature supervision for accurate machine-learning interatomic potentials. *arXiv:2603.04523*, 2025.
- A. Rodriguez, J. S. Smith, and J. L. Mendoza-Cortes. OpenREACT-CHON-EFH — Open REaction Dataset of Atomic ConfiguraTions comprising C, H, O, N with Energies, Forces, and Hessians. Figshare, doi:10.6084/m9.figshare.29189858, 2025.
- M. F. Hutchinson. A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines. *Commun. Stat. Simul. Comput.*, 19(2):433–450, 1990.
- B. A. Pearlmutter. Fast exact multiplication by the Hessian. *Neural Comput.*, 6(1):147–160, 1994.

- A. Griewank and A. Walther. *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*. SIAM, 2008.
- H. Avron and S. Toledo. Randomized algorithms for estimating the trace of an implicit symmetric positive semi-definite matrix. *J. ACM*, 58(2):1–34, 2011.
- A. K. Saibaba, A. Alexanderian, and I. C. F. Ipsen. Randomized matrix-free trace and log-determinant estimators. *Numer. Math.*, 137(2):353–395, 2017.
- N. Halko, P. G. Martinsson, and J. A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Rev.*, 53(2):217–288, 2011.
- D. P. Woodruff. Sketching as a tool for numerical linear algebra. *Found. Trends Theor. Comput. Sci.*, 10(1–2):1–157, 2014.
- R. Johnson and T. Zhang. Accelerating stochastic gradient descent using predictive variance reduction. *NeurIPS*, 26, 2013.
- R. H. Byrd, G. M. Chin, J. Nocedal, and Y. Wu. Sample size selection in optimization methods for machine learning. *Math. Program.*, 134(1):127–155, 2012.
- A. Loew, D. Sun, H.-C. Wang, S. Botti, and M. A. L. Marques. Universal machine learning interatomic potentials are ready for phonons. *npj Comput. Mater.*, 11(1):178, 2025.
- T. Cui, Y. Han, H. Jia, C. Duan, and Q. Zhao. A large scale molecular Hessian database for optimizing reactive machine learning interatomic potentials. *Sci. Data*, 13(1):37, 2026.
- T. Koker, A. Gangan, M. Kotak, J. Marian, and T. Smidt. PFT: Phonon fine-tuning for machine learned interatomic potentials. *arXiv:2601.07742*, 2026.
- A. Burger, L. Thiede, N. Rønne, V. Bernales, N. Vijaykumar, T. Vegge, A. Bhowmik, and A. Aspuru-Guzik. Shoot from the HIP: Hessian interatomic potentials without derivatives. *arXiv:2509.21624*, 2025.
- E. C.-Y. Yuan, A. Kumar, X. Guan, E. D. Hermes, A. S. Rosen, J. Zádor, T. Head-Gordon, and S. M. Blau. Analytical ab initio Hessian from a deep learning potential for transition state optimization. *Nat. Commun.*, 15(1):8865, 2024.

A Variance derivation for Proposition 2

Write the per-step Hessian gradient as $g = \frac{1}{B} \sum_i X_i$ with $X_i = \nabla_\theta |D_i v_i|^2$. Then

$$\text{Var}[g] = \frac{1}{B^2} \left[\sum_i \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right]. \quad (4)$$

Under independent probes ($K=B$) the cross terms vanish and $\text{Var}[g_{\text{PHL}}] = \frac{1}{B^2} \sum_i \text{Var}(X_i)$. Under PS-PHL, X_i and X_j are correlated only when $v_i = v_j$, i.e. $i \equiv j \pmod{K}$; such pairs number $\approx \binom{B}{2} \frac{K-1}{B-1} \approx \frac{B(K-1)}{2}$ out of $\binom{B}{2}$. For a Gaussian probe, $\text{Cov}(|D_i v|^2, |D_j v|^2) = 2\text{tr}(D_i^\top D_j D_j^\top D_i) + 2\text{tr}(D_i^\top D_i)\text{tr}(D_j^\top D_j)$, which is bounded by $\bar{\gamma} \text{Var}(|Dv|^2)$ with $\bar{\gamma}$ the trace coherence (2) when the per-molecule spectra are comparable. Substituting yields $\sigma_K^2 \leq \sigma_{\text{PHL}}^2 (1 + \bar{\gamma} c_K)$ with $c_K \leq 1$, and for $\bar{\gamma} \approx 0.03$ the inflation is $\lesssim 3\%$ regardless of K until K nears $3N_{\min}$, at which point the single-probe Rayleigh-quotient coverage of the smallest molecules breaks the isotropy that $\mathbb{E}[v_i v_i^\top] = I$ relies on for low-variance trace estimation.

B Reproducibility

Code, run logs (per-minibatch probe counts, per-epoch wall-clock, variance audits), evaluation scripts, and the aggregation/ t -test pipeline are in `proposal/code` and `proposal/logs`. Training used the authors’ patched TorchANI; seeds are pinned (`common.set_seed`); minibatch indexing and probe draws are logged for every run.